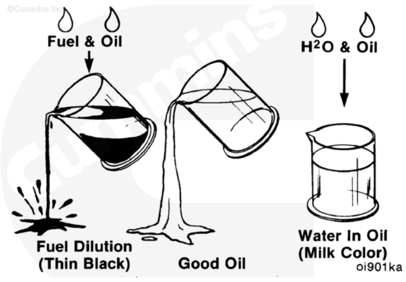
Trainings

General Information

Lubricating Oil Dilution

⚠ CAUTION ⚠

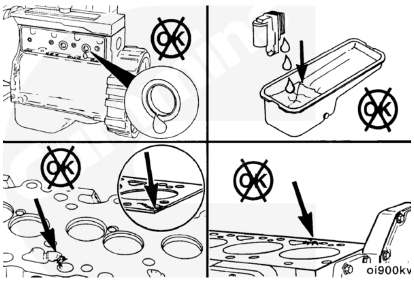
Diluted oil can cause severe engine damage.



Check the condition of the lubricating oil.

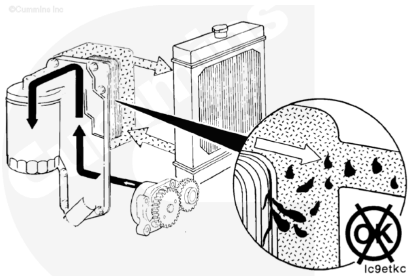
- Thin, black lubricating oil is an indication of fuel in the oil.
- Milky discoloration is an indication of coolant in the lubricating oil.

Coolant in the oil can be caused by:

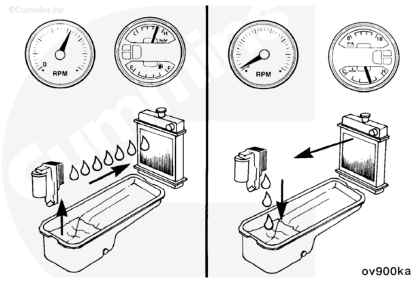


- Expansion plugs leaking
- Lubricating oil cooler element leaking
- Damaged cylinder head or gasket
- Cracked engine block
- Casting porosity.

Coolant-Diluted Lubricating Oil



Since the lubricating oil cooler design does**not** require gaskets or seals to maintain the separation of oil and coolant, the element itself **must** leak to allow mixing of the fluids. Refer to Procedure 007-003 (/qs3/pubsys2/xml/en/procedures/40/40-007-003-tr.html).

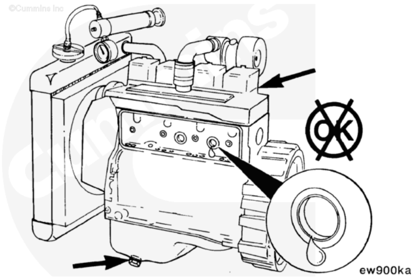


During operation, the lubricating oil pressure will be higher than coolant pressure. A leak in the lubricating oil cooler will show as lubricating oil in the coolant.

However, following an engine shutdown, the residual pressure in the coolant system can cause coolant to seep through the leak path into the lubricating oil.

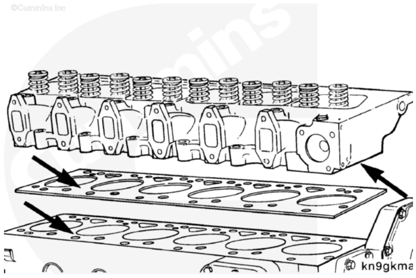
To check for leaks, pressurize the cooling system to 140 kPa [20 psi]. With the system pressurized, remove the following components, and inspect for leaks.

- Valve covers (leaks indicate cracked head)
- Lubricating oil drain plug (leaks indicate defective lubricating oil cooler, head gasket, cracked head or block)
- Tappet cover (expansion plug leak).



Coolant in the lubricating oil can be caused by a damaged cylinder head gasket or cracked cylinder head or block.

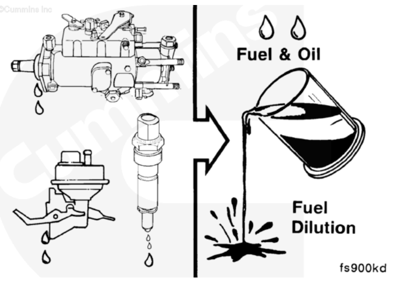
Remove the cylinder head and gasket, and inspect for cracks or damage.



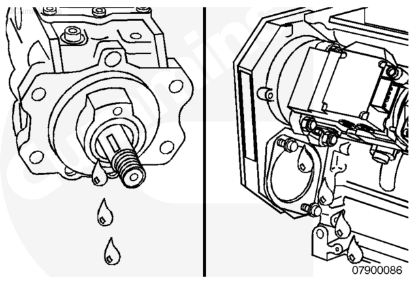
Fuel-Diluted Lubricating Oil

Fuel dilution is limited to five sources:

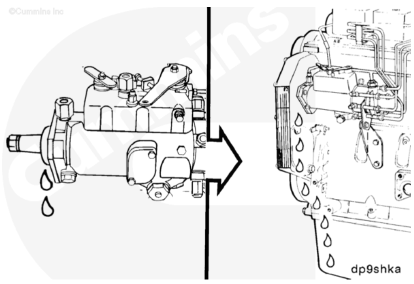
1. Injection pump shaft seal
2. Fuel leaking by the rings
3. Fuel transfer pump
4. A crack in the cylinder head from the fuel filter location to the air intake
5. Injector leakage.



Use the following logic to determine the source of the oil dilution with fuel:

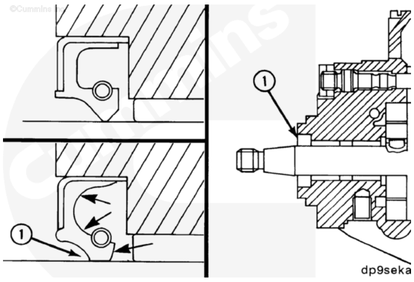


A worn or damaged fuel injection pump shaft seal will allow fuel to leak into the gear housing and then into the lubricating oil pan.

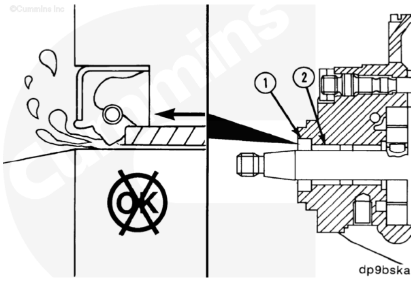


The seal is designed to provide increased sealing as the pump case pressure increases. Pressure forces the lip (1) tighter around the shaft.

A worn seal could leak during start-up and shutdown when case pressure is low. A worn seal can **not** easily be detected by pressurizing the pump.

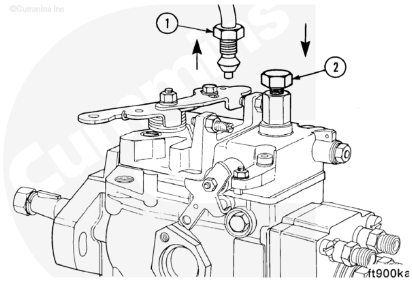


The bushing (2) in the Bosch® VE fuel injection pump can cause a seal leak. If the bushing is loose in the housing, it will move toward the seal raising the lip (1) and providing a leak path for fuel.

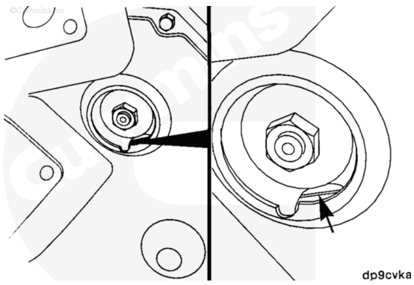


Note : The following steps only apply to front gear train engines. For rear gear train engines, there is no adequate access to check for fuel pump seal leakage on engine.

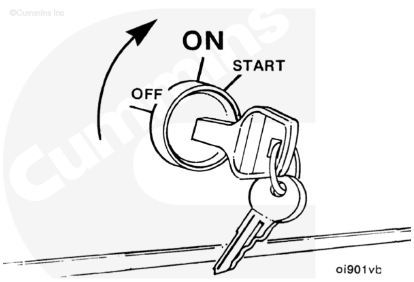
To check for such a leak, or a damaged seal (Bosch® VE **only**), remove the fuel drain manifold connection (1) at the pump, and install a plug (2).



Remove the access cover, and rotate the engine so one of the holes in the fuel injection pump gear exposes the back gear housing.

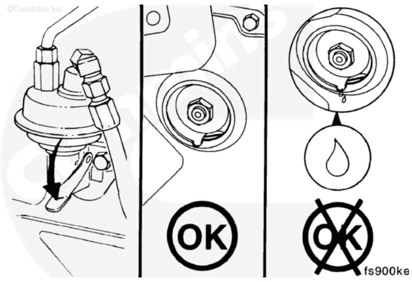


Activate the fuel shutdown valve by turning the switch to the ON position.



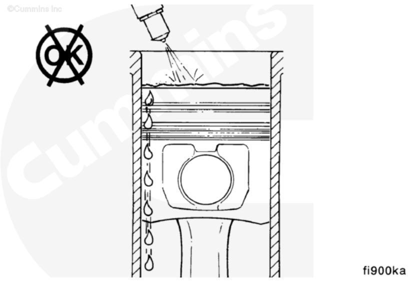
Use a small mirror to check for leaks while pumping the priming lever on the lift pump. If a leak is found, replace the injection pump. Refer to Procedure 005-014

(/qs3/pubsys2/xml/en/procedures/40/40-005-014-tr.html).



Incomplete combustion in the cylinders can result in unburned fuel draining into the oil pan.

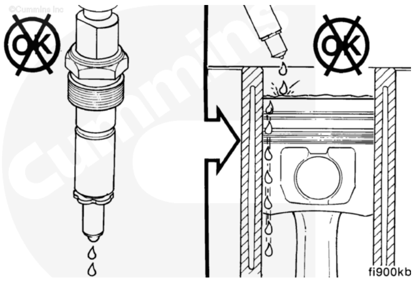
This condition can be caused by a leaking injector or reduced compression caused by inadequate piston ring sealing.



An increase in white exhaust smoke during the first start of the day is a symptom that an injector is leaking.

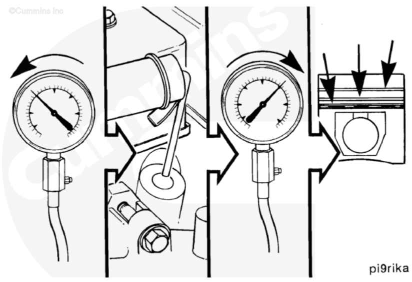
An injector leak will also cause the engine to run rough and have low power.

Remove and repair or replace leaking injectors. Refer to Procedure 006-026 (/qs3/pubsys2/xml/en/procedures/40/40-006-026-tr.html).



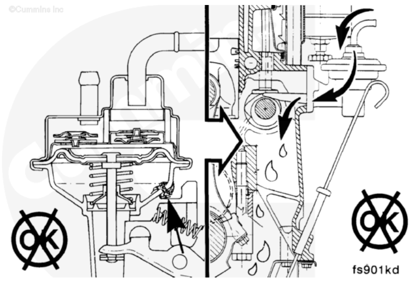
For two valve per cylinder engines, perform a compression check to verify piston ring sealing. Refer to Procedure 014-008 (/qs3/pubsys2/xml/en/procedures/40/40-014-008.html).

For four valve per cylinder engines, perform a blow-by check. Refer to Procedure 014-010 (/qs3/pubsys2/xml/en/procedures/100/100-014-010.html).

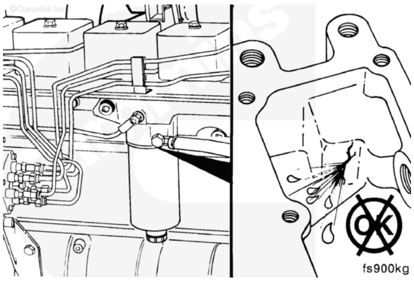


There is a remote possibility for fuel to drain into the oil from the diaphragm-type fuel transfer pump.

For this to happen, the diaphragm in the pump will break and the drain hole will need to be plugged.



On engines with head mounted fuel filters, another remote possibility, is that a crack or porosity in the head casting will allow fuel to leak to the air intake and onto the cylinders.



Last Modified: 31-Mar-2006